

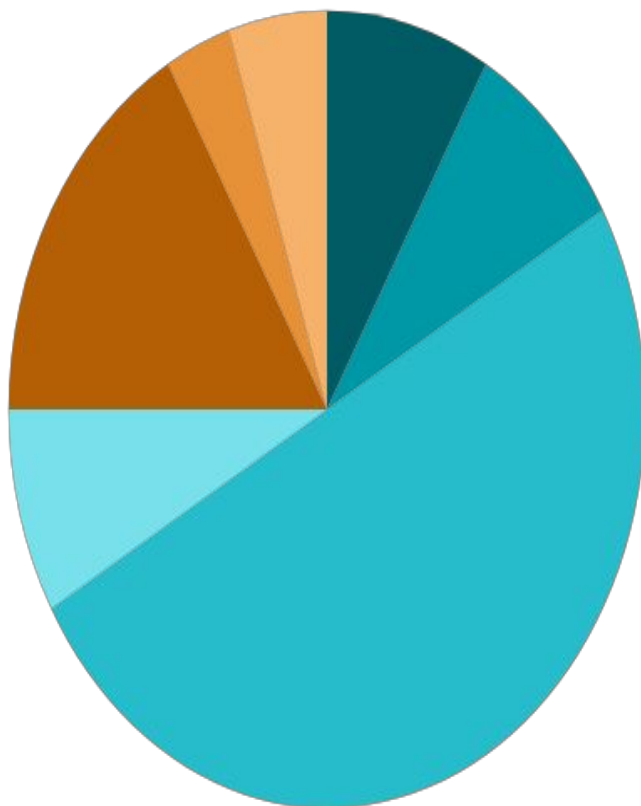


Introduction to Generative AI

Outline

- What is Artificial Intelligence, Machine Learning, and Deep Learning?
- Why Generative AI?
- Introduction to Generative Models
- Probability Distributions and Sampling Methods
- Types of Generative Models (GANs, VAEs, Diffusion Models)
- Applications of Generative AI
- Industry Examples and Case Studies
- Future Scope and Challenges of Generative AI

Lecture Structure



**05
Minutes**

Outline & Objectives



**05
Minutes**

Recap of Previous Lecture



**30
Minutes**

**Delivery of Actual Content as per
Lesson Plan**



**05
Minutes**

Industry Relevance



**10
Minutes**

Interactive Activities



**02
Minutes**

Importance of Discipline



**03
Minutes**

Recording of Attendance



Unit 1: Topics at a Glance

01

AI Overview

History, definitions, AI vs ML vs DL

02

ML & Deep Learning

Supervised, unsupervised, neural networks

03

Generative Models

Concepts, scope, discriminative vs generative

04

Probability

Gaussian, Multivariate, Sampling methods

05

GANs

Architecture, training, applications

06

VAEs

Latent space, ELBO, reparameterization

07

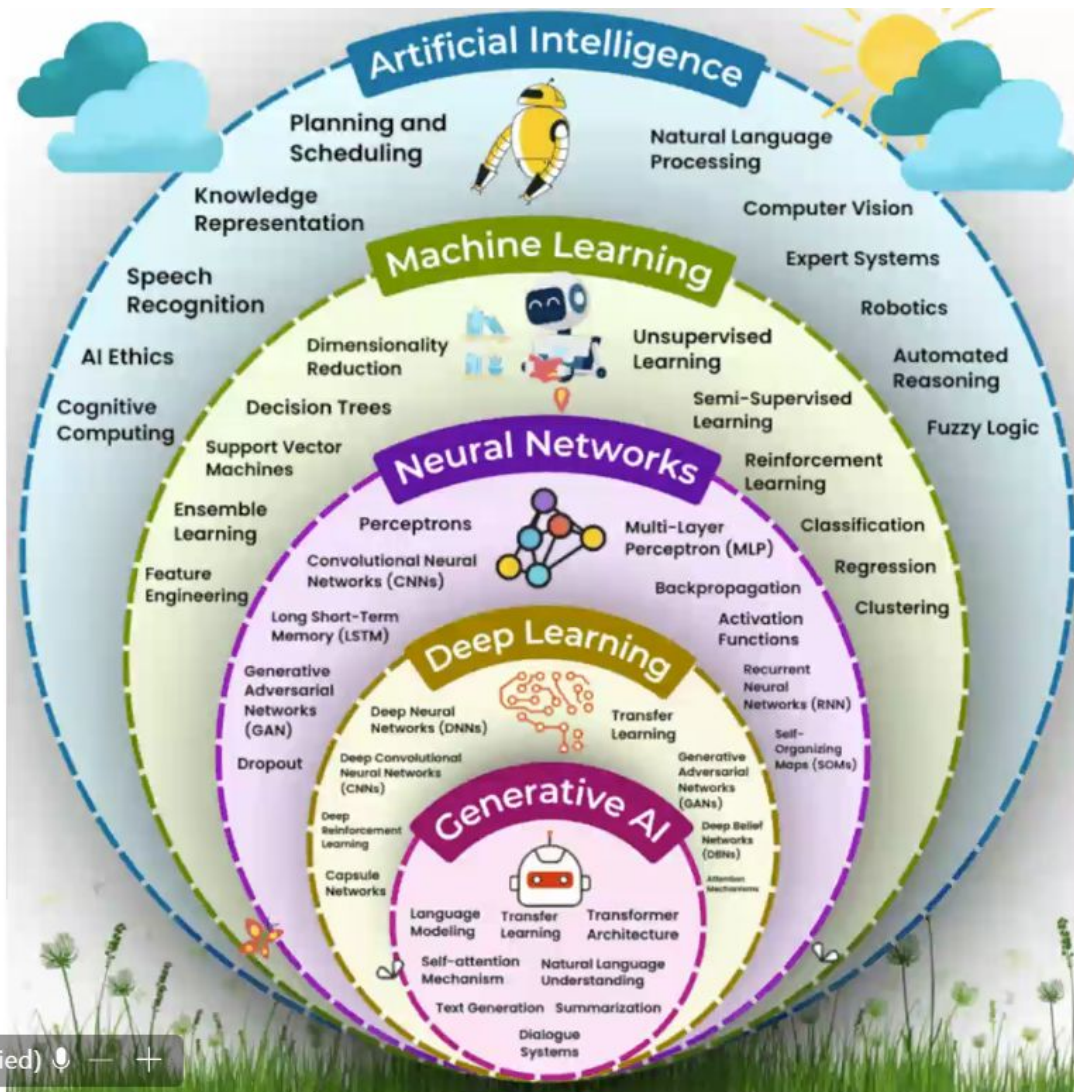
Diffusion Models

Forward/reverse, DDPM, score matching

08

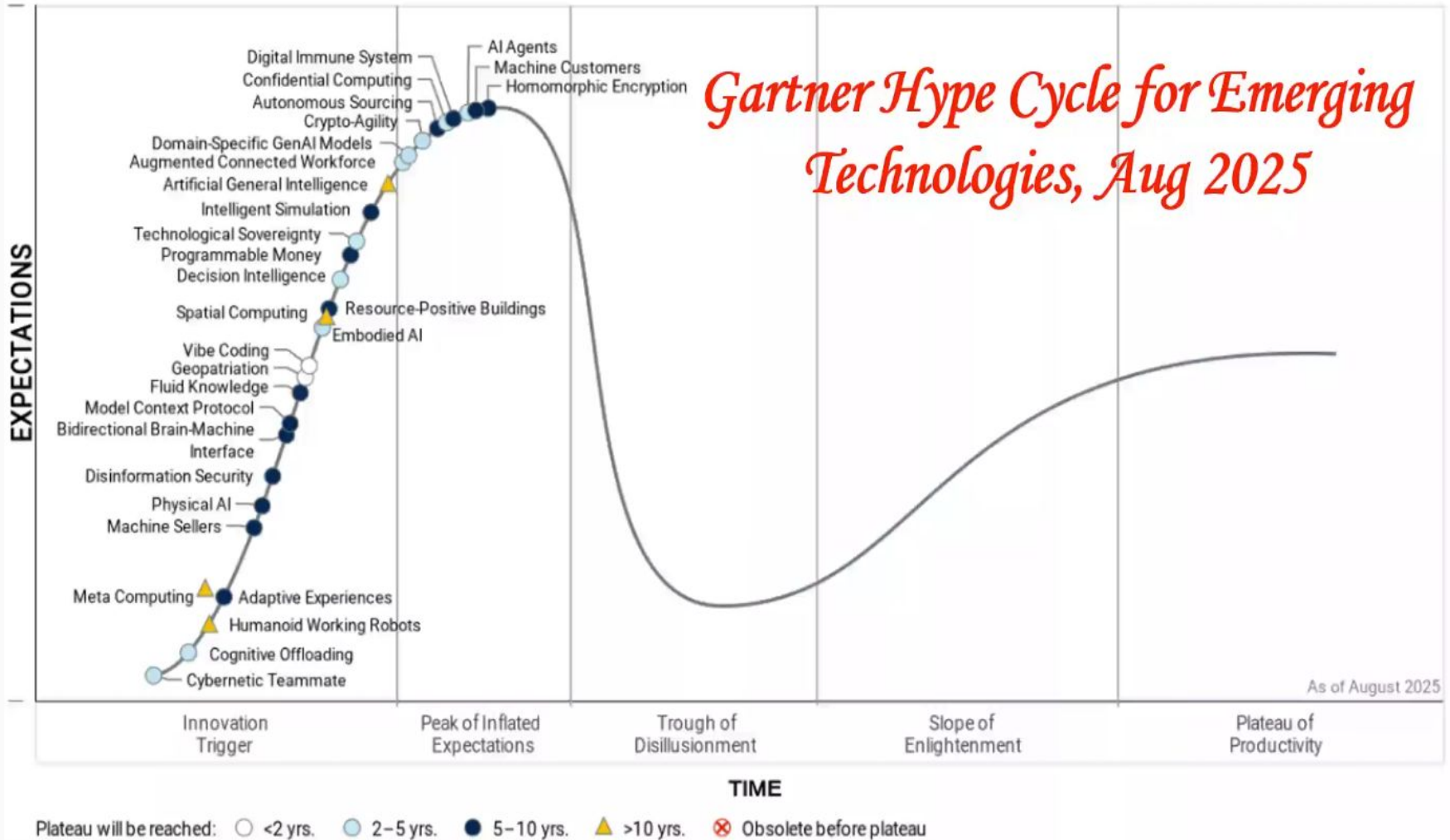
Applications

Image, text, drug discovery, music



The AI Universe

or (Unverified) 🔊 — +





The foundations of AI



Philosophy: Where does knowledge come from?



Linguistics: How does language relate to thought?



Neuroscience: How do our brains process information?



Behavioral Economics: How do you make decisions to maximize utility?



Mathematics: What can be computed?



Computer Science: How can we build an efficient computer?

SECTION 1

Overview of AI, Machine Learning & Deep Learning

From Turing's question to modern neural architectures

Introduction

Artificial Intelligence, Machine Learning and Deep Learning

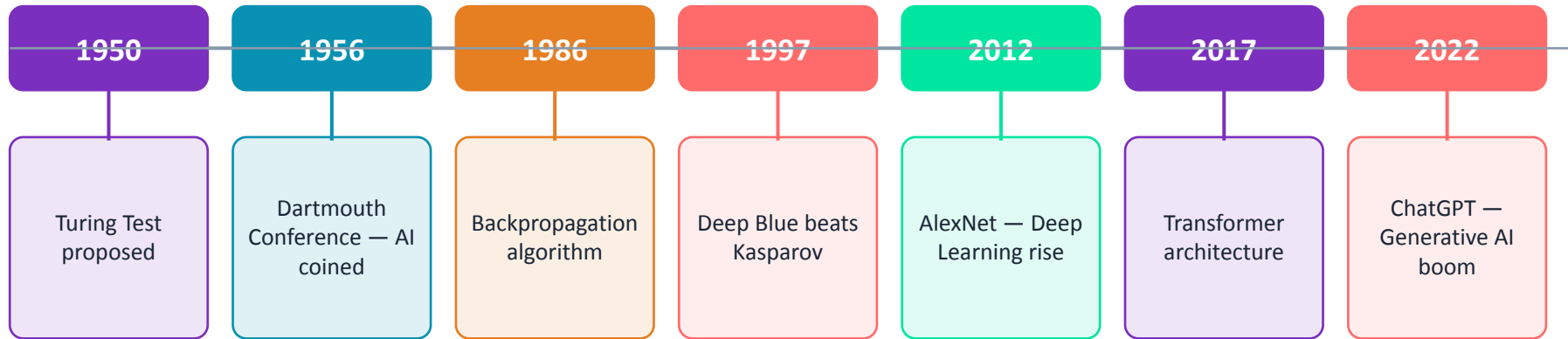
- ❑ The term **AI** in the broadest sense refers to simulation of human intelligence processes by computer systems
- ❑ **Machine Learning** is a subset of AI focusses on designing specific systems which can learn and make decisions/predictions based on data.
- ❑ **Deep Learning** is a subset of Machine Learning that uses a specific set of algorithms known as Neural-Networks often with many layers.



What is Artificial Intelligence?

- AI is the simulation of human intelligence processes by machines, especially computer systems
- Key processes: learning (acquiring information), reasoning (applying rules), self-correction
- Proposed by John McCarthy at Dartmouth Conference, 1956 — coined the term 'Artificial Intelligence'
- Alan Turing's seminal 1950 paper asked: 'Can machines think?' — introduced the Turing Test
- Modern definition: a system that perceives its environment and takes actions to maximize chance of success
- Types: Narrow AI (ANI) — task-specific; General AI (AGI) — human-level; Super AI (ASI) — beyond human
- Current state: We are in the era of Narrow AI (e.g., ChatGPT, AlphaGo, self-driving cars)

History & Milestones of Artificial Intelligence



 **AI Winters (1974–80, 1987–93):** Funding cuts due to unmet expectations — led to renewed focus on specialized systems and data-driven approaches



AI vs Machine Learning vs Deep Learning

Artificial Intelligence

- Rule-based & knowledge systems
- Expert systems, fuzzy logic

Machine Learning

- Learns from data patterns
- SVMs, Decision Trees, RF

Deep Learning

- Multi-layer neural nets
- CNNs, RNNs, Transformers
- **Generative AI lives here!**

What is Machine Learning? — Core Concepts

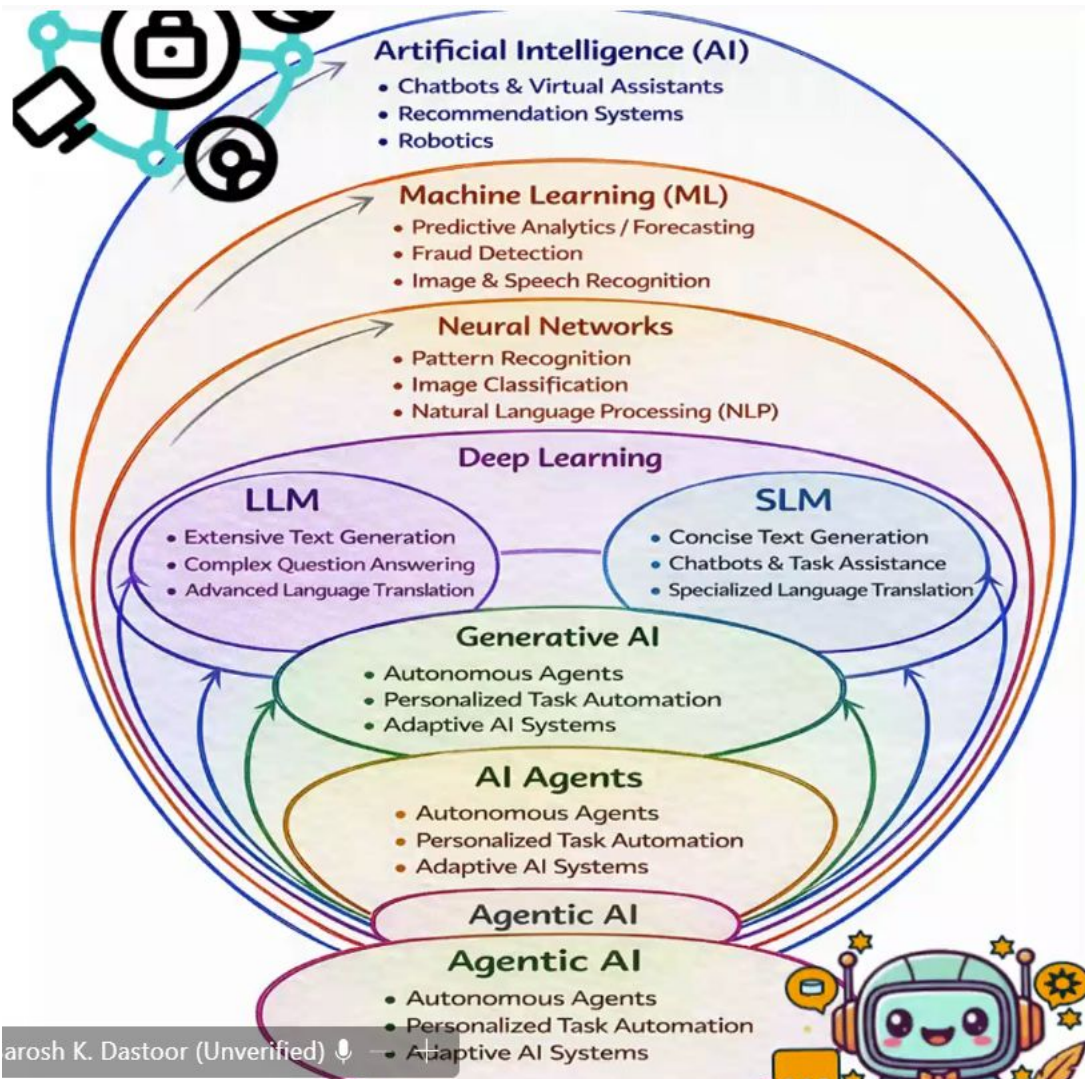


- Definition: Algorithms that improve automatically through experience (Tom Mitchell, 1997)
- Key idea: Instead of explicit programming, models learn patterns from data
- Three pillars: Data (fuel), Algorithm (engine), Compute (horsepower)
- Types:
 - Supervised: Labeled data → Classification & Regression (e.g., spam detection)
 - Unsupervised: Unlabeled data → Clustering & Dimensionality Reduction (e.g., customer segments)
 - Reinforcement: Agent learns via rewards/penalties (e.g., game-playing agents)
- ML pipeline: Data Collection → Preprocessing → Feature Eng. → Model Training → Evaluation → Deployment
- Generalization vs Overfitting: A model should perform well on unseen data, not just memorize training data
- Key metrics: Accuracy, Precision, Recall, F1-Score, MSE/RMSE for regression tasks

Deep Learning — Neural Network Fundamentals



- Deep Learning: Neural networks with multiple hidden layers that learn hierarchical representations
- Inspired by the human brain's interconnected neurons — but the analogy is loose!
- Artificial Neuron: Weighted sum of inputs $\rightarrow$ Activation function $\rightarrow$ Output
- Activation Functions: Sigmoid (0–1), ReLU ($\max(0, x)$), Tanh (–1 to 1), Softmax (probability distribution)
- Forward Pass: Input $\rightarrow$ Hidden layers $\rightarrow$ Output; Backpropagation: computes gradients using chain rule
- Gradient Descent: Update weights in direction that minimizes the loss function
- Key architectures: CNN (spatial/image), RNN/LSTM (sequential/text), Transformer (attention-based), GNN (graphs)
- Generative AI uses deep nets to learn the data distribution and sample new data from it



AI Ecosystem – Roles and responsibilities



GPT-4

~1.8 Trillion parameters

175B+

Parameters in GPT-3

~50x

GPU speedup vs CPU for DL

1000x

Data growth over last decade

ImageNet

14M images — key DL catalyst

2017

Transformer paper — 'Attention
Is All You Need'

ML Learning Paradigms — Detailed Breakdown

Supervised & Unsupervised Learning

- Supervised: Model maps input X to output Y using labeled pairs (x, y)
- Regression: Predicts continuous values — house prices, stock returns
- Classification: Predicts categories — spam/not spam, disease diagnosis
- Unsupervised: No labels — discover hidden structure in data
- Clustering: K-means, DBSCAN, Hierarchical clustering
- Dimensionality Reduction: PCA, t-SNE, UMAP for visualization
- Self-supervised: Model creates labels from data itself (e.g., masked language modeling in BERT)

Reinforcement & Transfer Learning

- Reinforcement Learning: Agent learns by interacting with environment
- Key components: State, Action, Reward, Policy, Value function
- Algorithms: Q-Learning, DQN, PPO, A3C — used in AlphaGo, ChatGPT RLHF
- Transfer Learning: Pretrain on large dataset, fine-tune on specific task
- Dramatically reduces data and compute requirements
- Examples: Fine-tuning GPT for legal documents, fine-tuning ResNet for medical imaging
- Foundation Models: Large pretrained models that serve as base for many tasks



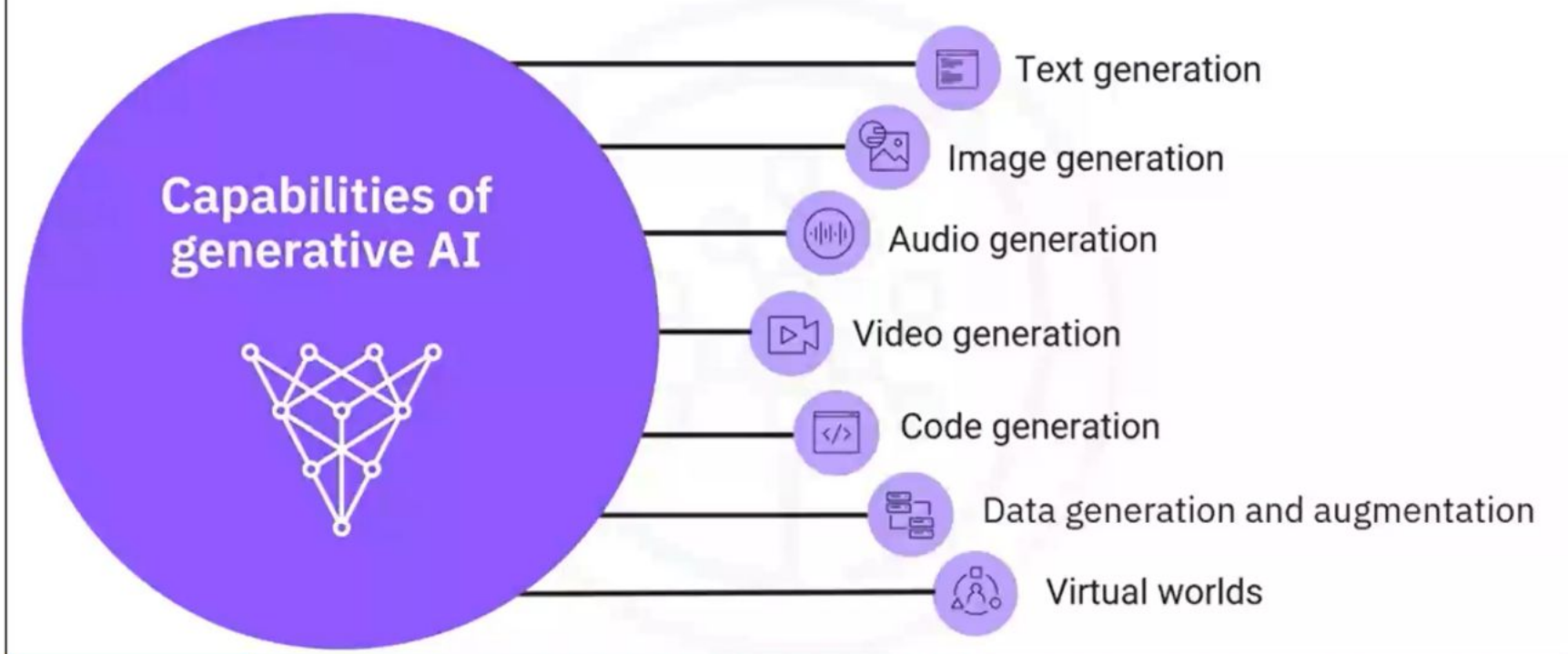
Introduction to Generative AI

Concepts, scope, and the shift from discrimination to generation

Generative AI is a kind of **high-end AI technology** that can produce numerous styles of content, together with text, imagery, audio and synthetic statistics.

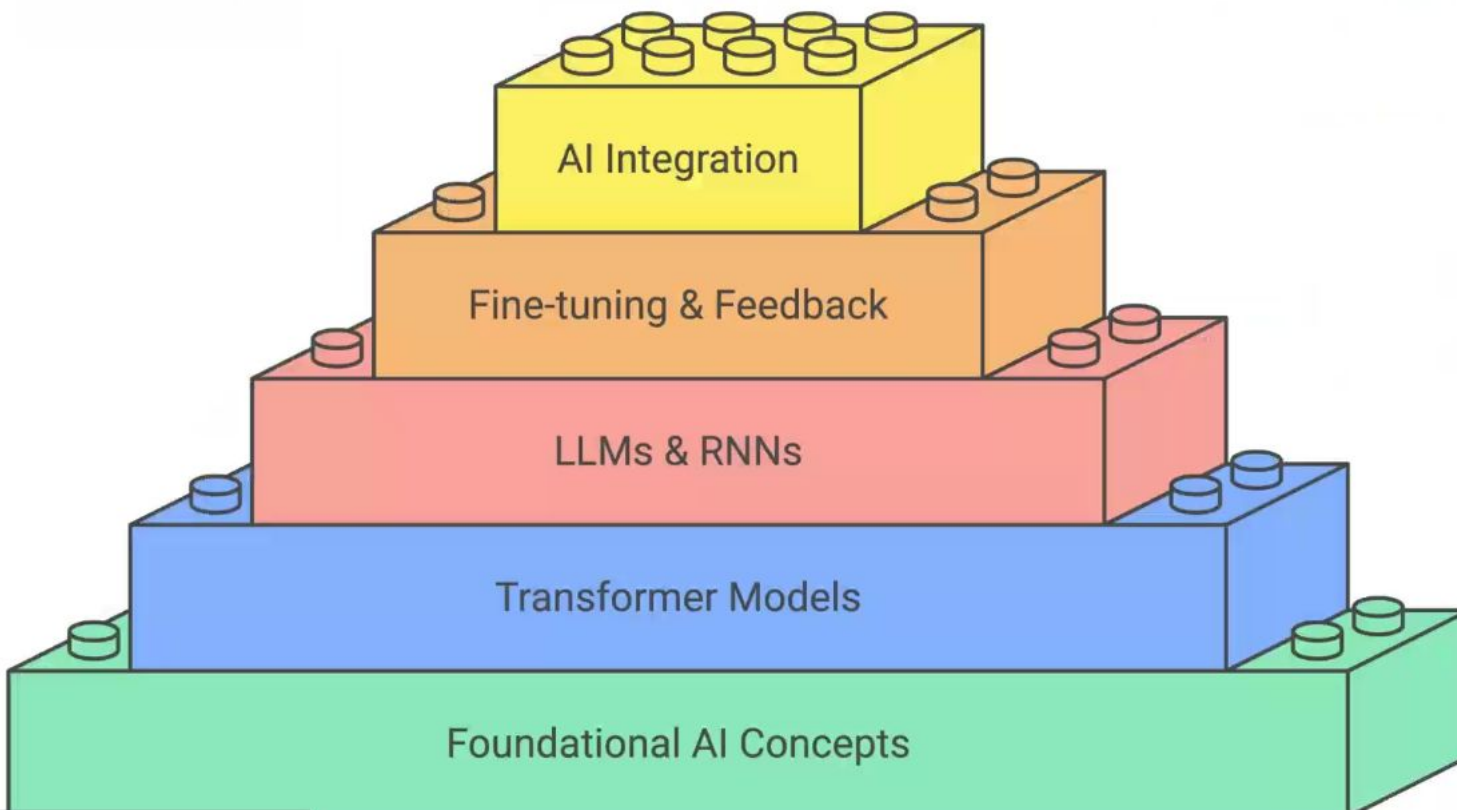
- **Prompt Engineering for Generative AI:** The art of asking the right question to get the best output from an LLM.
- Today, LLMs can be "programmed" in English, as well as [other languages](#).

Capabilities of generative AI





Hierarchy of Generative AI





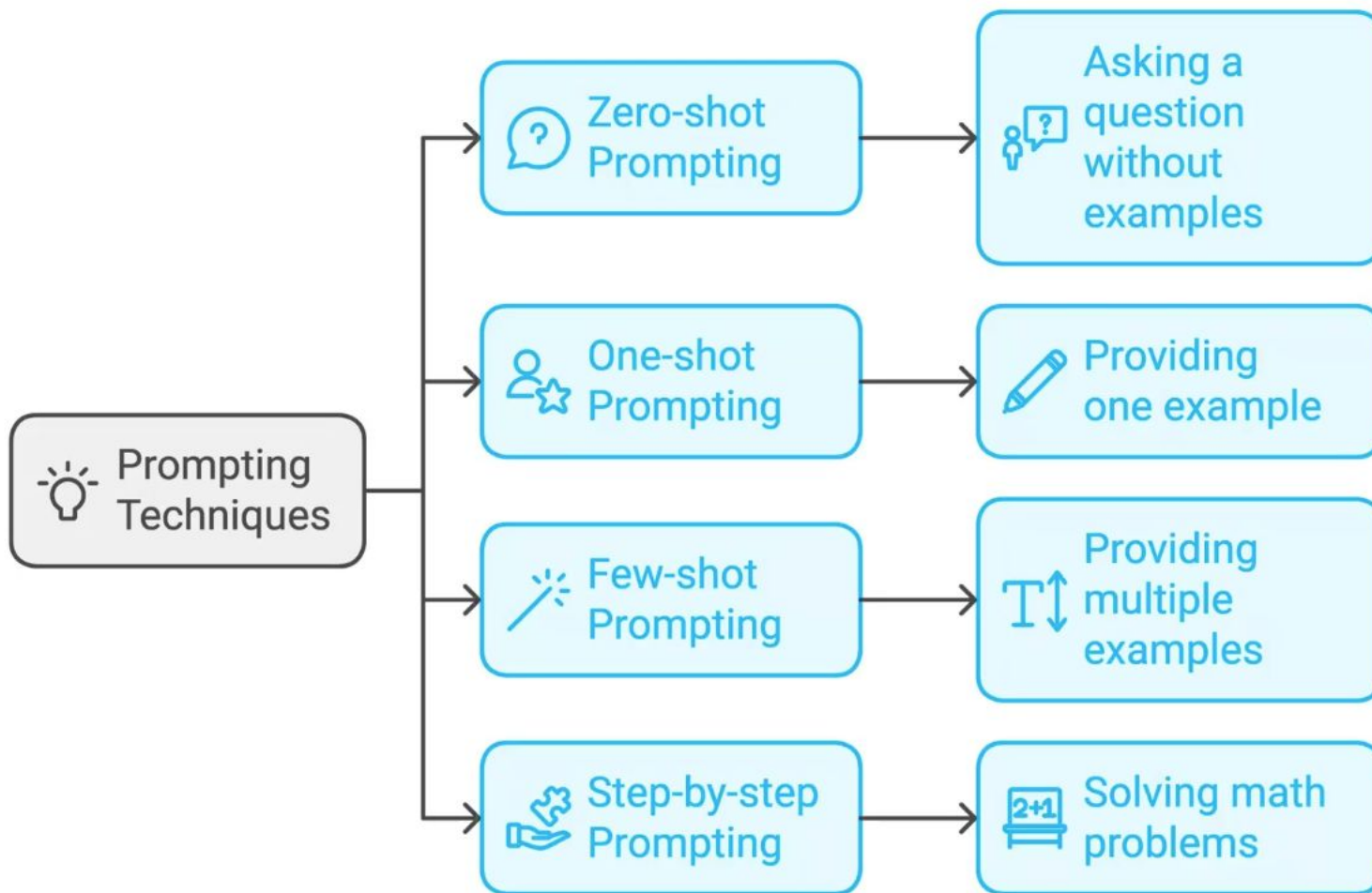
Generative AI – 12 Terms

Large Language Model

- Modeling human language at scale is a highly complex and resource-intensive endeavor.
- As models are built bigger and bigger, their complexity and efficacy increases.
- Modern large language models can predict the probability of sentences, paragraphs, or even entire documents.

How large is large?

*It is fuzzy, but "large" has been used to describe **BERT (110M parameters)** as well as **PaLM 2 (up to 340B parameters)**.*



Show conversation (unread messages)

Prompt Engineers Vs AI Engineers



ash K. Dastoor (Unverified)

Writing PERFECT Prompts using AI Assistant

The 5 Pillars of Perfect Prompts

1. BE SPECIFIC AND CLEAR

- **Bad:** "Tell me about marketing"
- **Good:** "Explain 5 digital marketing strategies for a small e-commerce business selling handmade jewellery, with budget considerations under \$500/month"

2. PROVIDE CONTEXT

- **Bad:** "Write a product description of eco-friendly bamboo toothbrush "
- **Good:** "Write a 150-word product description for an eco-friendly bamboo toothbrush. Target audience: environmentally conscious millennials. Tone: friendly and informative. Highlight: biodegradable, sustainable, plastic-free."

3. DEFINE THE FORMAT

- **Bad:** "Give me ideas for blog posts"
- **Good:** "Create a table with 10 blog post ideas for a fitness blog. Columns: Title, Target Audience, Main Benefit, Estimated Word Count"



4. ASSIGN A ROLE

- **Bad:** "Help me with my resume"
- **Good:** "Act as an experienced HR manager with 15 years of experience. Review my software engineer resume and provide 5 specific improvements to make it stand out for senior positions at tech companies."

5. USE STEP-BY-STEP INSTRUCTIONS

- **Bad:** "Analyze this data"
- **Good:** Analyze this sales data step by step:
 1. Identify the top 3 performing products
 2. Calculate month-over-month growth rate
 3. Highlight any unusual patterns or anomalies
 4. Provide 3 actionable recommendations
 5. Present findings in bullet points with numbers

The evolution of Intelligence

Era	Question
Computing	Can machines calculate?
AI	Can machines think?
Machine Learning	Can machines learn?
Deep Learning	Can machines recognize?
Generative AI	Can machines create?

The Chef analogy

- Traditional AI
- A food critic. Looks at food and says:
- Pizza
- Pasta
- Burger
- **Classification.**
- Generative AI
- A chef.
- Creates a completely new recipe.
- **Generation.**



Types of AI Models

Discriminative
technique



Classify

Discriminative model
(classify as a dog or a cat)



DOG

Generative
technique



Generate

Generative model
(generate dog image)



Meet the New Creators

Generative AI: The New Creators

• Text. Images. Video. Music. All Generated by AI. •

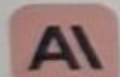
TEXT



ChatGPT



Gemini



Claude

IMAGES



Midjourney



Stable Diffusion

VIDEO



Sora



Veo

MUSIC

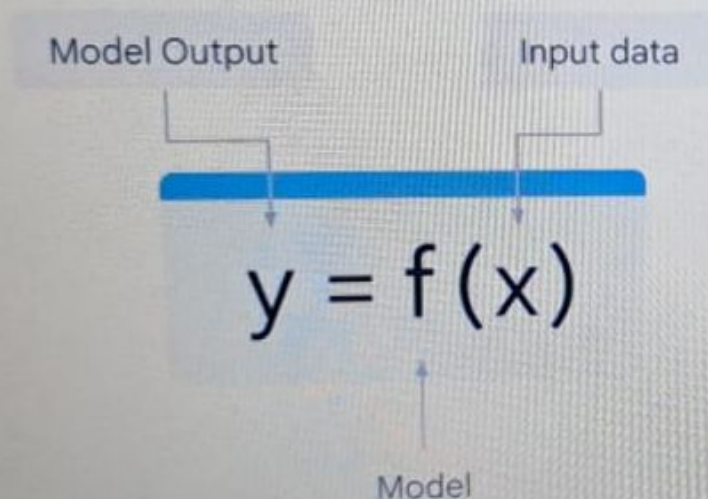


Suno



Udio

When Gen AI and when Not



Not GenAI when y is a:

- Number
- Discrete
- Class
- Probability

Is GenAI when y is:

- Natural language
- Image
- Audio

Generative Models

Generative **language** models

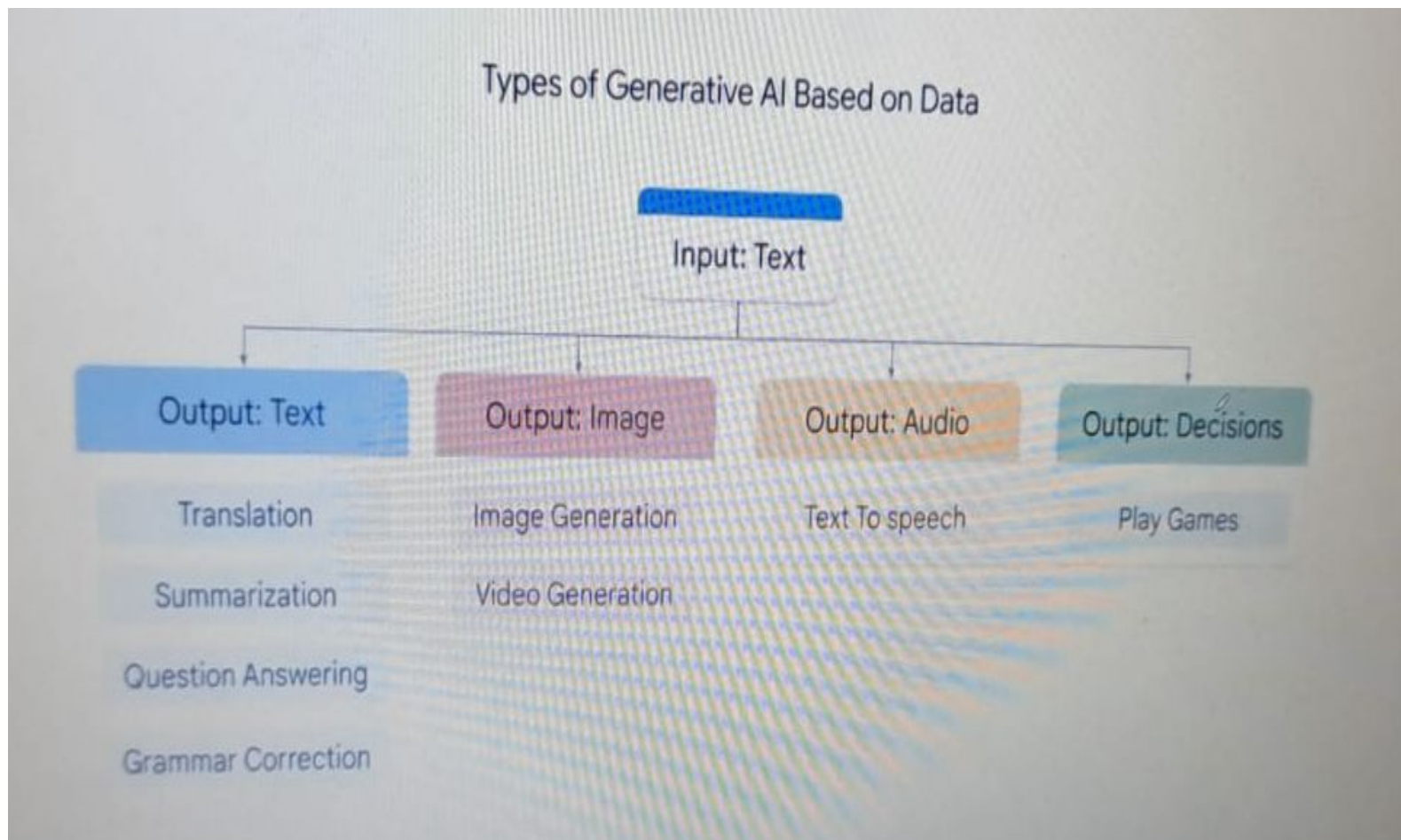
Generative language models learn about patterns in language through training data.

Then, given some text, they predict what comes next.

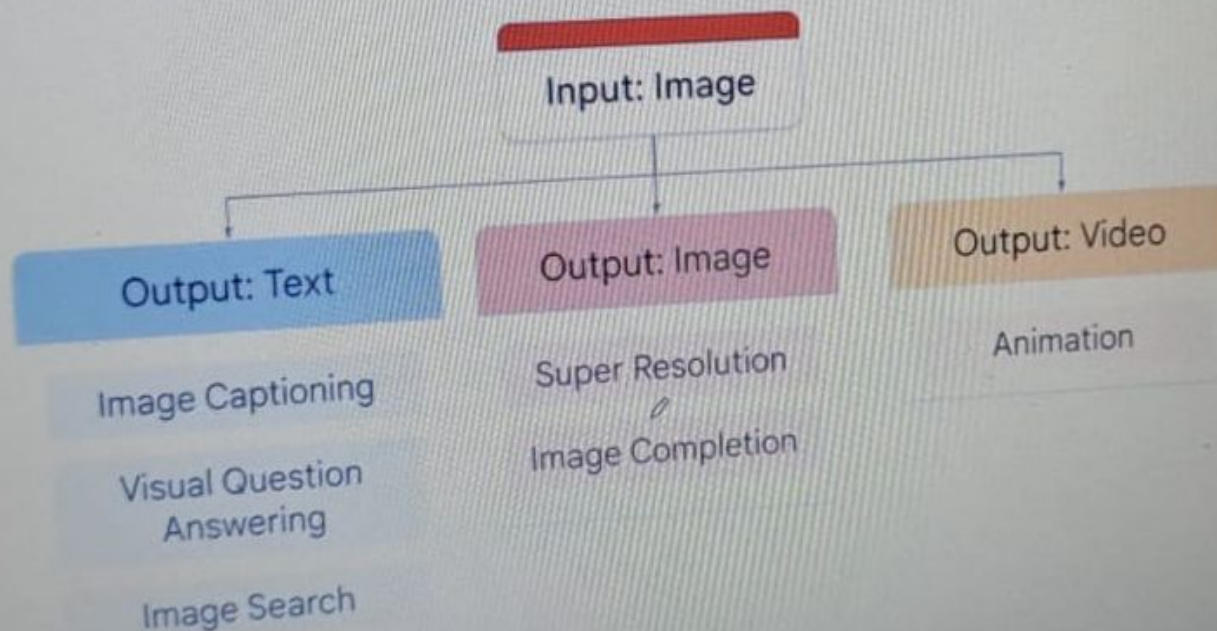
Generative **image** models

Generative image models produce new images using techniques like diffusion.

Then, given a prompt or related imagery, they transform random noise into images or generate images from prompts.



Types of Generative AI Based on Data



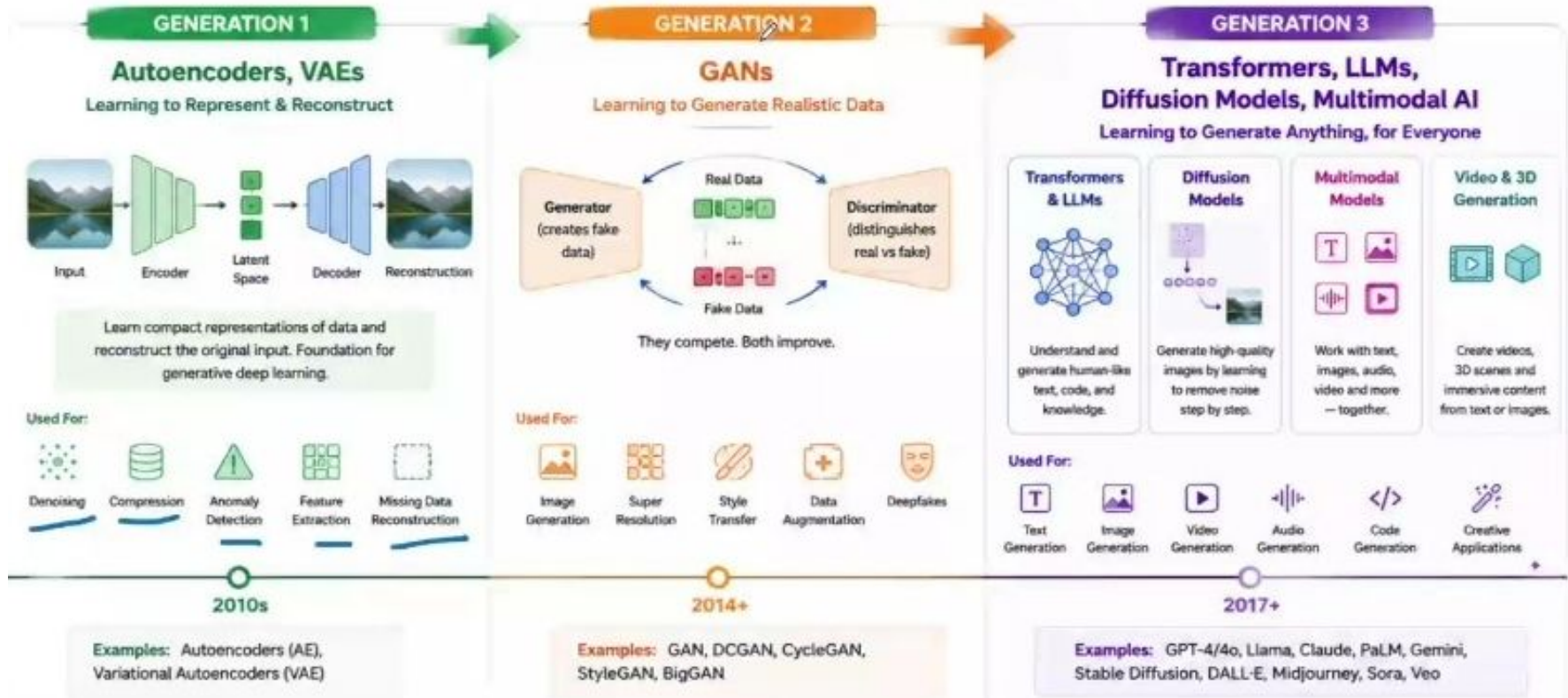


Introduction to Generative Models

Concepts, scope, and the shift from discrimination to generation

Evolution of Generative Models

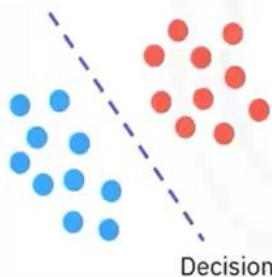
From Learning Representations to Creating Anything



Discriminative AI

Distinguishes between different classes of data

Each data point labeled with its class



Use advanced algorithms to:

- Differentiate
- Classify
- Identify patterns
- Draw conclusions

Example: Email spam filters

Generative AI

Prompt

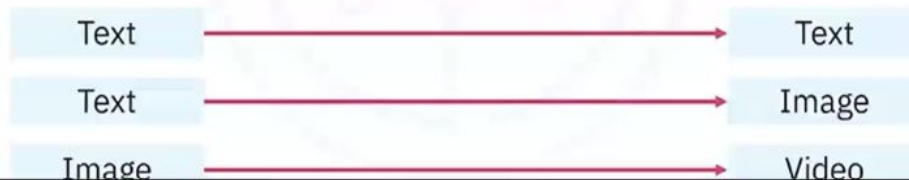
- Text
- Image
- Video
- Other forms of input



Output

- Text
- Images
- Audio
- Video
- Code
- Data

Generative AI



Discriminative vs Generative Models

Discriminative Models

- Learn $P(Y|X)$ — probability of label given input
- Focus: Decision boundary between classes
- Examples: Logistic Regression, SVM, CNN classifiers, BERT for classification
- Task: 'Is this email spam or not?' — given email, predict class
- Pros: Often higher accuracy on classification tasks with enough labeled data
- Cons: Cannot generate new samples from the data distribution
- Cannot answer: 'What does a typical spam email look like?'

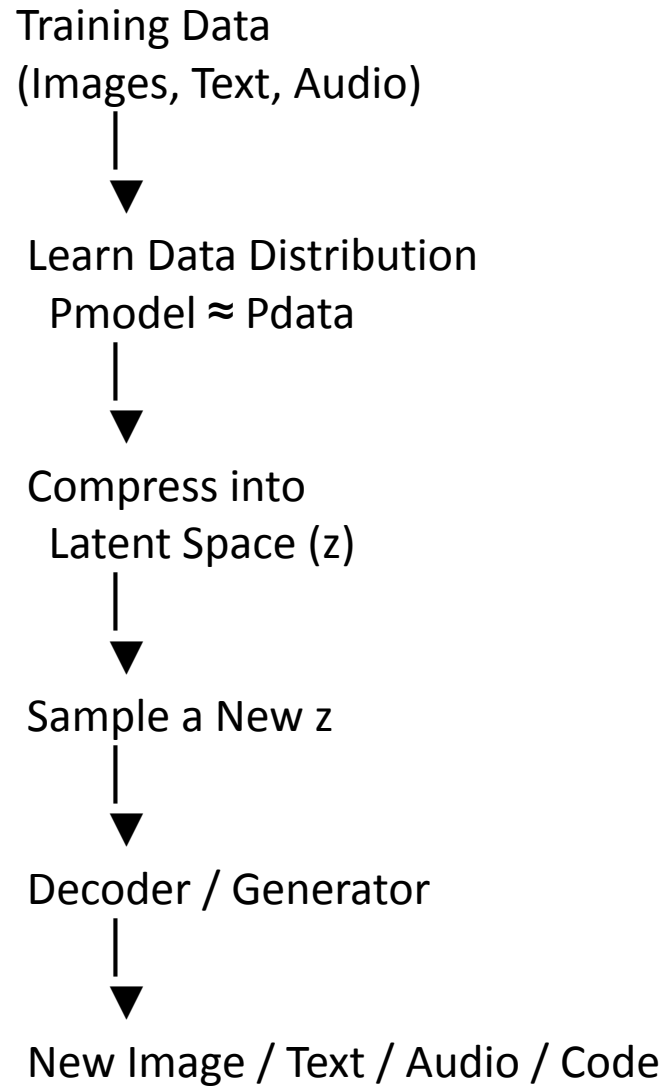
Generative Models

- Learn $P(X)$ or $P(X,Y)$ — the full joint/marginal distribution
- Focus: Understand how data is generated
- Examples: GANs, VAEs, Diffusion Models, GPT series, Stable Diffusion
- Task: Generate a realistic image / realistic text passage from scratch
- Pros: Can generate new data, learn rich representations, perform density estimation
- Cons: Harder to train, mode collapse, evaluation is difficult
- Key question: What is the 'true' underlying distribution of natural images/text?



What Are Generative Models? — Core Idea

- Generative models learn the underlying probability distribution of training data
- Goal: Given training data X drawn from true distribution $P_{data}(x)$, learn a model $P_{model}(x) \approx P_{data}(x)$
- Once learned, sample $x_{new} \sim P_{model}(x)$ to generate new, realistic data points
- The Challenge: P_{data} is extremely complex — for images, it's a distribution over $224 \times 224 \times 3 = 150,528$ dimensions!
- Latent Space: A lower-dimensional, compressed representation where structure is learned
- Concept: 'z' (latent vector) encodes 'concepts' — e.g., z_1 = smile, z_2 = age, z_3 = lighting
- Applications span: Images, text, audio, molecules, protein structures, code, 3D shapes
- The dream: A generative model that understands the world well enough to create any content on demand





\$1.3T

Projected GenAI market by 2032
(McKinsey)

200M+

ChatGPT users as of 2024

4.4B

AI images generated daily (2024
est.)

95%

Text readable as human-written
(GenAI)

10×

Drug discovery acceleration w/
GenAI

30%+

Code written with AI assistance
(GitHub survey)



Mathematical Foundation of Generative Models



- Probability Distribution: A function describing likelihood of each possible value of a random variable
- Joint Distribution $P(X,Y)$: Probability of both X and Y occurring simultaneously
- Marginal Distribution $P(X) = \sum_y P(X,Y)$ — summing out unwanted variables
- Bayes' Theorem: $P(Y|X) = P(X|Y) \cdot P(Y) / P(X)$ — fundamental to probabilistic modeling
- Log-Likelihood: $\log P(X|\theta)$ — we maximize this during training (numerical stability)
- KL Divergence: $D_{KL}(P||Q) = \sum P(x) \log[P(x)/Q(x)]$ — measures distance between distributions
- Maximum Likelihood Estimation (MLE): Find $\theta^* = \operatorname{argmax}_{\theta} \sum \log P(x_i|\theta)$ — standard training objective
- Evidence Lower Bound (ELBO): Used in VAEs to make optimization tractable

Taxonomy of Generative Models

Explicit Density Models

- Model $P(x)$ directly and exactly
- Autoregressive Models (PixelCNN, GPT)
- Normalizing Flows (exact likelihood)
- Tractable density
- Slow generation (sequential)

Implicit Density Models

- Learn to sample without explicit $P(x)$
- Generative Adversarial Networks (GANs)
- Score-based / Diffusion Models
- No explicit likelihood
- Often high-quality outputs

VAEs: Approximate Density (ELBO)
Bridge between explicit & implicit



Probability Distributions



Gaussian, Multivariate, and Sampling Methods



Why Probability Theory is Central to Generative AI

Suppose

Exam Marks

45

52

60

63

67

70

72

73

75

80

Notice Most students score around 70.

Very few score 40 or 95.

This pattern is called Probability Distribution.



Why Probability Theory is Central to Generative AI

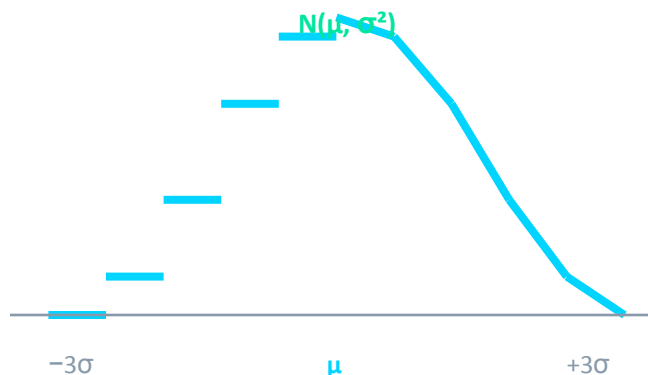
- All generative models are probabilistic — they model distributions over complex data
- A Gaussian image model assumes pixels follow normal distributions (often wrong but useful!)
- Training = finding distribution parameters that best explain the training data
- Sampling = drawing new examples from the learned distribution to generate content
- Latent variable models (VAEs, Diffusion) rely on structured probability distributions
- KL Divergence (measure between distributions) appears in VAE loss and information theory
- Monte Carlo Sampling: Estimating intractable integrals by averaging random samples
- Reparameterization trick (VAEs): Makes gradient flow through stochastic nodes possible

Gaussian (Normal) Distribution — Deep Dive

$$\text{PDF: } f(\mathbf{x}) = \frac{1}{\sigma \sqrt{2\pi}} \times \exp\left(-\frac{(\mathbf{x}-\mu)^2}{2\sigma^2}\right)$$

μ = Mean (location) σ = Std Dev (spread) σ^2 = Variance

- Bell-shaped, symmetric about mean μ
- 68-95-99.7 rule: $\pm 1\sigma$, $\pm 2\sigma$, $\pm 3\sigma$ cover 68%, 95%, 99.7% of data
- Standard Normal: $\mu=0$, $\sigma=1 \rightarrow N(0,1)$ — used in VAE prior
- Central Limit Theorem: Sum of i.i.d. random variables $\rightarrow$ Gaussian
- Fully specified by just 2 parameters: μ and σ (elegant!)
- Used in: VAE prior $p(z)=N(0,I)$, noise in diffusion models





Multivariate Gaussian Distribution

- Extension of 1D Gaussian to d-dimensional random vector $x \in \mathbb{R}^d$
- Parameterized by:
 - Mean vector $\mu \in \mathbb{R}^d$ (d-dimensional center)
 - Covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ (shape and orientation of distribution)
- PDF: $f(x) = (1/(2\pi)^{d/2} |\Sigma|^{1/2}) \times \exp(-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu))$
- Diagonal Σ : Variables are independent — simplifies to product of 1D Gaussians
- Full Σ : Captures correlations between dimensions (e.g., height $\leftrightarrow$ weight)
- In VAEs: encoder outputs μ and $\log(\sigma^2)$ for each latent dimension (diagonal covariance assumed)
- Contour plots of MVN are ellipses — eigenvectors of Σ give principal axes
- Mahalanobis distance: $(x-\mu)^T \Sigma^{-1} (x-\mu)$ — accounts for scale and correlation

Sampling Methods — Generating from Distributions

Direct Sampling Methods

- Inverse Transform Sampling: Generate $u \sim \text{Uniform}(0,1)$, apply $F^{-1}(u)$ to get sample
- Box-Muller Transform: Generate two uniforms $\rightarrow$ transform into two standard normals
- Ancestral Sampling: In graphical models, sample variables in topological order
- Importance Sampling: Sample from proposal $q(x)$, weight by $p(x)/q(x)$
- Used when direct sampling from $p(x)$ is hard but evaluating $p(x)$ is easy
- Monte Carlo Estimation: Approximate $E[f(x)] \approx (1/N) \sum f(x_i)$ where $x_i \sim p(x)$

Markov Chain Monte Carlo (MCMC)

- MCMC: Construct a Markov chain whose stationary distribution is $p(x)$
- Metropolis-Hastings: Propose $x' \sim q(x'|x)$, accept with prob $\min(1, \frac{p(x')q(x|x')}{p(x)q(x'|x)})$
- Gibbs Sampling: Update each variable conditioned on all others — used in early generative models
- Langevin Dynamics: $x_{t+1} = x_t + \eta \nabla_x \log p(x) + \sqrt{(2\eta)\epsilon}$ — connects to score-based models!
- HMC (Hamiltonian Monte Carlo): Uses gradient information for efficient exploration
- MCMC is the foundation of Diffusion Model's reverse process understanding

Sampling in Practice — GenAI Inference



- Ancestral Sampling in Autoregressive Models (GPT): $x_1 \sim p(x_1)$, then $x_2 \sim p(x_2 | x_1)$, ..., $x_{\square} \sim p(x_{\square} | x_1, \dots, x_{\square-1})$
- Temperature Sampling: Divide logits by T before softmax — $T=1$ (standard), $T<1$ (sharper/repetitive), $T>1$ (more random)
- Top-k Sampling: Sample from only the k most likely tokens — reduces gibberish
- Nucleus (Top-p) Sampling: Sample from smallest vocab set with cumulative probability $\geq p$ (adaptive)
- Reparameterization Trick (VAEs): $z = \mu + \sigma \odot \epsilon$ where $\epsilon \sim N(0, I)$ — makes sampling differentiable!
- DDPM Sampling (Diffusion): Iterative denoising over T steps — $x_T \sim N(0, I)$, denoise to x_0
- DDIM Sampling: Deterministic faster version of DDPM — 10-50 steps vs 1000
- Classifier-Free Guidance: Balance diversity and quality via guidance scale w



Generative AI

$N(0,1)$

Standard Gaussian — VAE prior
& diffusion noise

Bernoulli

Binary data — pixel on/off in
bitmaps

Categorical

Discrete tokens — language
models (softmax output)

Laplace

Sparse priors, robust models

$\text{Beta}(\alpha, \beta)$

Probability priors — bounded
[0,1] outputs

Dirichlet

Topic models — distribution over
distributions

Generative Adversarial Networks (GANs)

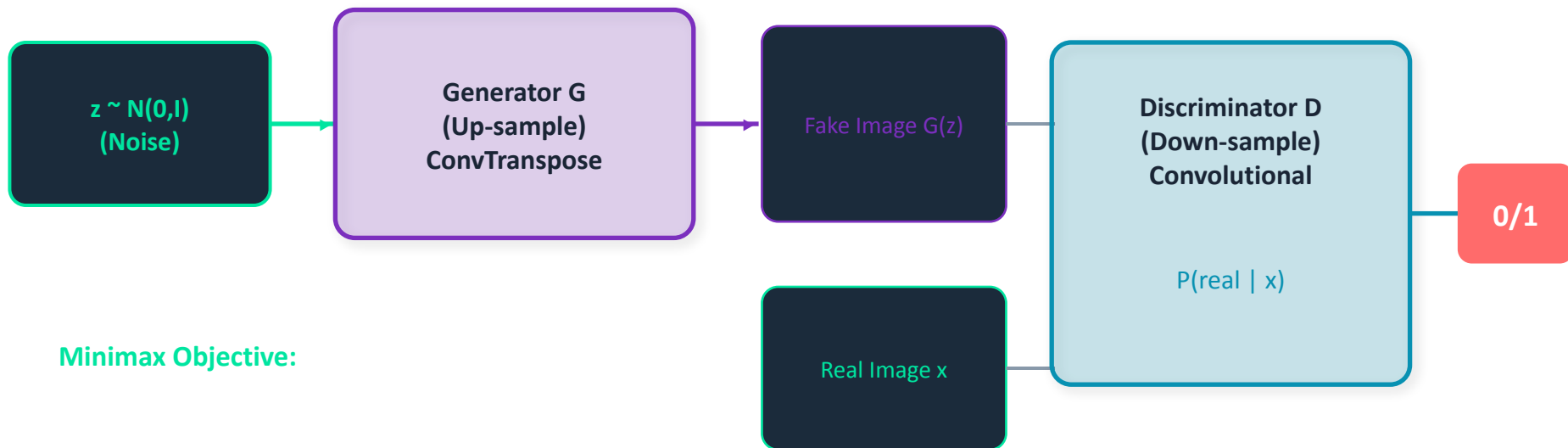
The adversarial game that revolutionized image synthesis

Generative Adversarial Networks — Overview



- Introduced by Ian Goodfellow et al. in 2014 — one of the most influential ideas in ML
- Core concept: Two neural networks compete in a zero-sum game (adversarial training)
- Generator G: Takes random noise $z \sim p(z)$ and generates fake data $G(z)$
- Discriminator D: Takes real or fake data, outputs probability of being real
- G wants to fool D; D wants to catch G — they improve each other through competition
- Analogy: G is a counterfeiter, D is a detective — both get better as they compete
- At convergence (Nash equilibrium): G produces data indistinguishable from real data
- No explicit likelihood: GANs are implicit density models — we sample but can't evaluate $P(x)$

GAN Architecture — Generator & Discriminator



Minimax Objective:

$$\min_G \max_D V(D, G) = E_{\{x \sim p_{\text{data}}\}} [\log D(x)] + E_{\{z \sim p_z\}} [\log (1 - D(G(z)))]$$

D maximizes: distinguishing real from fake | G minimizes: making D output 1 for fakes



GAN Training Process — Step by Step

- Step 1: Sample mini-batch of real images x from training data
- Step 2: Sample noise $z \sim N(0, I)$, generate fake images $G(z)$
- Step 3: Train Discriminator:
 - $L_D = -[\log D(x) + \log(1 - D(G(z)))]$ — maximize D's accuracy
- Step 4: Fix D, train Generator:
 - $L_G = -\log D(G(z))$ (non-saturating variant) — maximize D's probability of labeling fakes as real
- Alternate Steps 3 & 4 for each mini-batch iteration
- Training tricks: Label smoothing, feature matching, gradient penalty, spectral normalization
- Mode Collapse: G generates only a few modes of data — major training failure
- Convergence indicator: D's accuracy hovers near 50%; FID (Fréchet Inception Distance) decreases

Major GAN Variants & Their Innovations

Architecture Variants

- DCGAN (2015): Deep Convolutional GAN — stable training with ConvNet
- Conditional GAN (cGAN): G and D receive class label — generate specific categories
- Progressive GAN (ProGAN): Grow resolution gradually — 1024×1024 faces
- StyleGAN (2019): Style-based generator — control hair, age, expression separately
- StyleGAN2/3: Fixes artifacts — used in 'This Person Does Not Exist'
- BigGAN: Large-scale class-conditional — ImageNet quality
- CycleGAN: Unpaired image-to-image translation — horse ↔ zebra

Loss Function Variants

- Vanilla GAN: BCE loss — training instability due to vanishing gradients
- LSGAN (Least Squares GAN): L2 loss — more stable gradients
- WGAN (2017): Wasserstein distance — solves mode collapse, better convergence
- WGAN-GP: Gradient Penalty for WGAN — enforces Lipschitz constraint
- Spectral Normalization GAN (SNGAN): Normalizes weight matrices
- InfoGAN: Maximizes mutual information — learns disentangled representations
- Self-Attention GAN (SAGAN): Attention for long-range spatial dependencies

GAN Applications — Real-World Impact



- Image Synthesis: High-resolution photorealistic face generation (StyleGAN) — 'This Person Does Not Exist'
- Image-to-Image Translation: pix2pix — satellite $\leftrightarrow$ map, sketch $\leftrightarrow$ photo, day $\leftrightarrow$ night
- Super Resolution: SRGAN, ESRGAN — upscale low-res images with realistic detail
- Video Generation: VideoGAN, MoCoGAN — temporal coherent video synthesis
- Medical Imaging: Data augmentation for rare conditions, synthetic CT/MRI generation
- Text-to-Image (early): AttnGAN, StackGAN — text description $\rightarrow$ image (precursor to DALL-E)
- Domain Adaptation: CycleGAN for unpaired translation — no aligned training pairs needed
- Drug Discovery: Molecule generation with MolGAN — design molecules with desired properties



Mode Collapse

Generator produces only a few output types

Training Instability

Oscillating loss — difficult to converge

FID Score

Fréchet Inception Distance — standard quality metric

Nash Equilib.

Theoretical convergence not guaranteed in practice

Eval. Gap

High FID $\neq$ human perception of quality

No $P(x)$

Cannot compute exact likelihood — limits applications



Variational Autoencoders (VAEs)

Structured latent spaces and tractable generation

From Autoencoders to Variational Autoencoders

Standard Autoencoder (AE)

- Architecture: Encoder $\rightarrow$ Bottleneck $\rightarrow$ Decoder
- Encoder: Compresses input $x \rightarrow$ latent code z (lower dimension)
- Decoder: Reconstructs $\hat{x}$ from z ; minimizes $\|x - \hat{x}\|^2$
- Learning: Unsupervised — learns compact representations without labels
- Latent space: NOT structured — similar inputs may map to scattered points
- Problem for generation: Random z points in latent space may decode to garbage
- Use case: Compression, denoising, anomaly detection — not generation!

Variational Autoencoder (VAE)

- Key innovation: Encoder outputs distribution params μ, σ instead of point z
- Encoder maps $x \rightarrow (\mu, \sigma)$ — parameters of a Gaussian $q(z|x)$
- Sample $z \sim N(\mu, \sigma)$ using reparameterization trick: $z = \mu + \sigma \odot \epsilon$
- Decoder: $p(x|z)$ — reconstructs from sampled z
- KL term forces posterior $q(z|x)$ to stay close to prior $N(0, I)$
- Result: Smooth, structured latent space — interpolation works!
- Generation: Sample $z \sim N(0, I)$, decode $\rightarrow$ new realistic sample

VAE Mathematics — Evidence Lower Bound (ELBO)

Goal: Maximize $\log P(x) = \log \int P(x|z) P(z) dz$ (Intractable!)

$$\text{ELBO} = \underbrace{E_{q(z|x)}[\log P(x|z)]}_{\text{Reconstruction Term}} - \underbrace{KL(q(z|x) \parallel P(z))}_{\text{Regularization Term}}$$

Reconstruction:
 $E[\log P(x|z)]$ — measures how well decoder reconstructs input
 → MSE for images, BCE for binary

KL Regularization:
 $KL(N(\mu, \sigma^2) \parallel N(0, I))$
 $= \frac{1}{2} \sum (\mu^2 + \sigma^2 - \log \sigma^2 - 1)$
 → Forces latent space structure

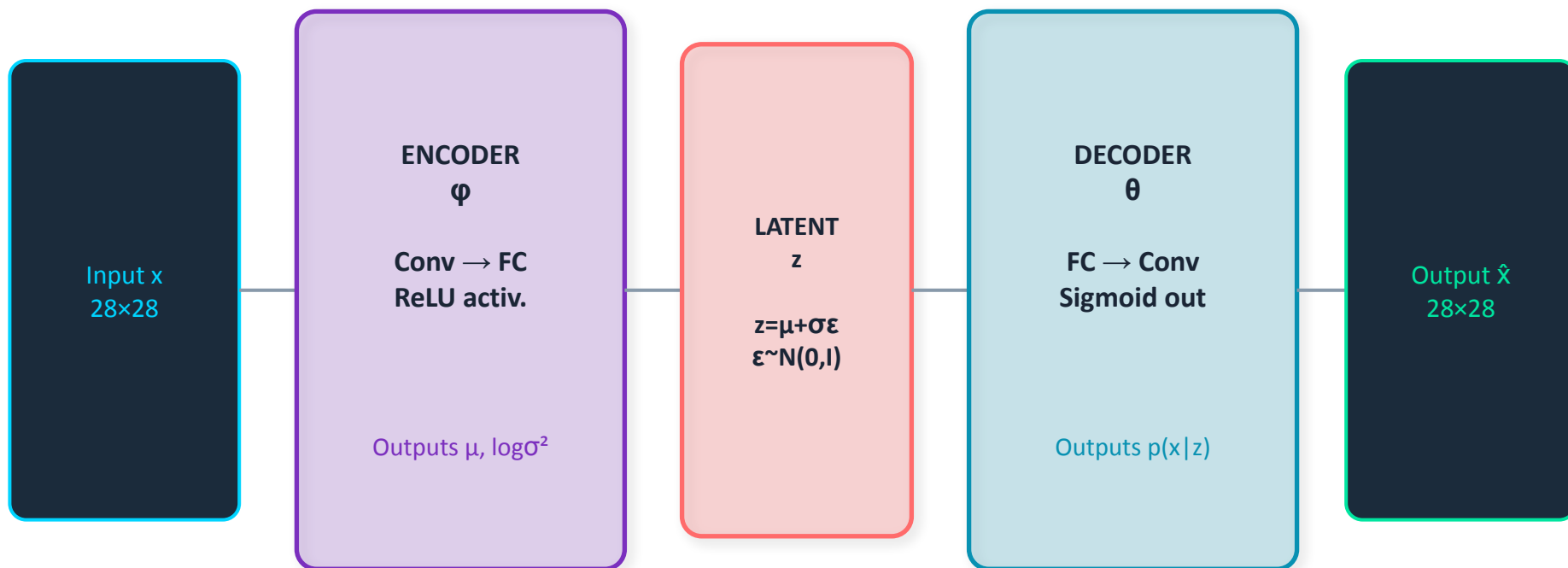


Reparameterization Trick — Making Sampling Differentiable



- Problem: Backpropagation cannot flow through a stochastic sampling node
- We need $\partial L / \partial \mu$ and $\partial L / \partial \sigma$, but $z \sim N(\mu, \sigma^2)$ is a random operation — no gradient!
- Reparameterization: $z = \mu + \sigma \odot \epsilon$, where $\epsilon \sim N(0, I)$ is auxiliary noise
- Key insight: Randomness is now in ϵ (not a parameter), so gradients can flow through μ and σ
- Forward pass: ϵ sampled $\rightarrow z$ computed as deterministic function of $\mu, \sigma \rightarrow$ loss computed
- Backward pass: $\partial z / \partial \mu = 1$, $\partial z / \partial \sigma = \epsilon$ — standard backpropagation works!
- This trick enabled the entire VAE framework — without it, training would be impossible
- Generalizes to other distributions: Gumbel-Softmax trick for discrete latent variables

VAE Architecture — Encoder, Latent Space & Decoder



Loss = Reconstruction Loss + $\beta \times$ KL Divergence (β -VAE uses $\beta > 1$ for better disentanglement)

VAE Latent Space — Interpolation & Disentanglement



- Smooth Interpolation: Because KL term enforces continuity, walking between two z points gives smooth transitions
- Example: Interpolate z between 'smiling face' and 'neutral face' — gradual transition in output space
- Disentanglement: Each latent dimension ideally controls one independent factor of variation
- β -VAE (Higgins et al., 2017): Increasing KL weight β encourages more disentangled representations
- In practice: z_1 might control smile, $z_2 \rightarrow$ rotation, $z_3 \rightarrow$ lighting (not always this clean)
- Conditional VAE (CVAE): Condition on label $c \rightarrow$ control what category is generated
- VQ-VAE: Discrete latent variables using vector quantization — used in DALL-E v1 image tokenizer
- Hierarchical VAE (HVAE/NVAE): Multiple levels of latent variables capture multi-scale structure

VAE vs GAN — Comparative Analysis

Aspect	VAE	GAN
Training	Stable (single loss)	Unstable (adversarial)
Objective	Maximize ELBO	Minimax game
Sample Quality	Blurry (MSE bias)	Sharp, photorealistic
Latent Space	Structured, smooth	Unstructured
Interpolation	Excellent	Unreliable
Likelihood	Approximate (ELBO)	None (implicit)
Mode Coverage	Good (avoids collapse)	Mode collapse risk
Inference Speed	Fast (single pass)	Fast (Generator only)
Best Use Case	Representation learning	High-quality synthesis



Diffusion Models

State-of-the-art generative models powering modern AI art



Diffusion Models — Core Intuition

- Inspired by non-equilibrium thermodynamics — diffusion of molecules in physics
- Key idea: Gradually add noise to data until it becomes pure Gaussian noise, then learn to reverse this process
- Forward process q : Real image $x_0 \rightarrow$ slowly corrupted $x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_T \sim N(0, I)$
- Reverse process p_θ : Learn a neural network to denoise $x_T \rightarrow x_{T-1} \rightarrow \dots \rightarrow x_0$
- At each step, the model predicts the noise that was added (or the clean image) and removes it
- Think of it as: Start with TV static, gradually reveal the image through T denoising steps
- Key insight: Each individual denoising step is a simple problem — denoise slightly corrupted image
- DDPM (2020, Ho et al.): First highly successful diffusion model — matched GAN quality on CIFAR-10

Diffusion Models — Forward & Reverse Process Math



- Forward Process (Markov chain): $q(x_t | x_{t-1}) = N(x_t; \sqrt{1-\beta_t} \cdot x_{t-1}, \beta_t \cdot I)$
- β_t is the noise schedule (small $\rightarrow$ large): typically linear or cosine schedule
- Closed-form at any t : $x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1-\bar{\alpha}_t} \cdot \epsilon$, where $\epsilon \sim N(0, I)$, $\bar{\alpha}_t = \prod_{s=1}^t (1-\beta_s)$
- This closed form lets us jump to any noise level in one step — no iteration needed during training!
- Reverse Process: $p_\theta(x_{t-1} | x_t) = N(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$
- Neural Network ϵ_θ predicts the noise ϵ from noisy image x_t at step t
- Training loss: $L = E_{t, x_0, \epsilon} [\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} \epsilon, t)\|^2]$ — simple MSE on noise!
- Inference: Start from $x_T \sim N(0, I)$, run learned reverse process T steps to get x_0

Diffusion Models — U-Net Backbone Architecture



- U-Net: The standard neural architecture used as ϵ_θ in DDPM and subsequent models
- Encoder (Downsampling): Series of ResBlocks + downsampling $\rightarrow$ extracts multi-scale features
- Bottleneck: Self-attention layers for global context — captures long-range dependencies
- Decoder (Upsampling): Mirrors encoder with skip connections from encoder $\rightarrow$ preserves detail
- Skip Connections: Direct paths from encoder to decoder — crucial for reconstruction quality
- Time Embedding: Sinusoidal positional encoding of t added to each ResBlock — tells model 'how noisy'
- Cross-Attention: In text-conditioned models, text embeddings injected via cross-attention layers
- Modern variant: DiT (Diffusion Transformer) — replaces U-Net with Vision Transformer blocks (FLUX, SD3)

Guided Diffusion & Accelerated Sampling

Classifier & Classifier-Free Guidance

- Text-to-image: Need to condition on text prompt — 'a cat in space'
- Classifier Guidance: Use pretrained classifier gradients to steer generation
- Classifier-Free Guidance (CFG): Train model with and without conditioning
- CFG formula: $\hat{\epsilon} = \epsilon_{\text{uncond}} + w \cdot (\epsilon_{\text{cond}} - \epsilon_{\text{uncond}})$
- w (guidance scale): Higher = more faithful to prompt, less diverse
- $w=1$: Unconditional, $w=7.5$: Good balance, $w=20+$: Overly saturated
- Used in: DALL-E 2, Stable Diffusion, Midjourney, Imagen

DDIM — Fast Sampling

- DDPM requires $T=1000$ steps — too slow for practical use
- DDIM (2020): Deterministic sampling, skips steps — 50 steps $\approx$ 1000 step quality
- DDIM inverts sampling — encode image to latent, edit, decode
- DPM-Solver: Adaptive step size solver — 20 steps comparable to DDPM-1000
- LCM (Latent Consistency Model): 4-8 steps — near real-time generation
- Turbo models (SDXL-Turbo, SD-Turbo): Adversarial distillation — 1-4 steps
- Flow Matching: Straighter trajectories (FLUX, SD3) — even fewer steps needed

Latent Diffusion Models — Stable Diffusion Architecture



- Key insight: Running diffusion in pixel space is slow and expensive ($256 \times 256 \times 3 = 196\text{K}$ dims)
- Solution: First encode image to compressed latent space using VAE, then run diffusion in latent space
- Compression ratio: Stable Diffusion uses $8\times$ spatial compression — $512 \times 512 \rightarrow 64 \times 64 \times 4$
- Architecture: VAE Encoder $\rightarrow$ Latent Space Diffusion (U-Net with attention) $\rightarrow$ VAE Decoder
- Text conditioning: CLIP/T5 text encoder $\rightarrow$ text embeddings $\rightarrow$ cross-attention in U-Net
- ControlNet: Add structural conditioning (edge maps, depth, pose) — preserves spatial structure
- SDXL: Larger 2.3B parameter model, native 1024×1024 , improved quality
- FLUX: Flow Matching + Transformers (DiT) — state of the art as of 2024



ry

FID 1.79

DDPM on CIFAR-10 — matches
best GANs

T=1000

DDPM reverse steps — DDIM
reduces to ~50

4.4B

FLUX.1 parameters — largest
open diffusion model

CFG w

Guidance scale: creativity vs
fidelity control

Stable

Training: no adversarial
instability (vs GANs)

ELBO?

Diffusion has exact ELBO unlike
GANs — tractable



Key Applications of Generative AI

Image synthesis, text generation, drug discovery, and beyond

Application 1: Image Synthesis & Creative AI



- Text-to-Image: DALL-E 3, Midjourney v6, Stable Diffusion, Adobe Firefly — 'Generate a surreal painting of a cat astronaut'
- Image Editing: Inpainting (fill masked regions), outpainting (extend canvas), style transfer
- Super-Resolution: ESRGAN, Real-ESRGAN — 4× upscale with hallucinated details
- Image-to-Image: ControlNet, IP-Adapter — transform sketch to photo, change style
- 3D Generation: DreamFusion, Zero-1-to-3 — generate 3D models from text or single image
- Video Generation: Sora, RunwayML Gen-3, Stable Video Diffusion — text/image to video
- Industry Impact: Marketing, game art, concept design, fashion design, architecture visualization
- Revenue: Adobe Stock AI images, Getty Images licensing, Canva AI — billion-dollar market

Application 2: Text Generation & Large Language Models



- Foundation: Transformer + Autoregressive modeling — GPT, LLaMA, Gemini, Claude family
- GPT-4: 1.8T+ parameters, multimodal, ~100 page context window — state of the art (2024)
- Training: Pretraining on internet text (self-supervised) → RLHF alignment for helpfulness/safety
- Capabilities: Writing, coding, summarization, translation, Q&A, reasoning, mathematics
- Code Generation: GitHub Copilot, Cursor, Claude Code — 30%+ of enterprise code uses AI assistance
- RAG (Retrieval-Augmented Generation): Combines LLMs with vector databases for factual grounding
- Agents: LLMs with tool use (search, calculator, browser) — autonomous task execution
- Challenges: Hallucinations, context limits, bias, prompt injection, copyright, factuality



Application 3: Drug Discovery & Molecular Generation

- Traditional drug discovery: 10-15 years, \$2.6 billion per drug — 90% failure rate in clinical trials
- GenAI acceleration target: Design molecules with specific target properties in days, not years
- Molecular Representations: SMILES strings (text), molecular graphs (2D/3D), 3D point clouds
- Generative Models for Molecules:
 - VAEs on SMILES (e.g., Molecular VAE by Gómez-Bombarelli)
 - GAN-based: MolGAN on molecular graphs
 - Diffusion: DiffSBDD, TargetDiff — structure-based drug design
- AlphaFold 2 (DeepMind): Solved protein structure prediction — 200M protein structures predicted
- Property Optimization: Reinforce to optimize drug-likeness (QED), synthesizability (SA), binding affinity
- Real-world: Insilico Medicine's INS018_055 — first AI-designed drug to reach Phase 2 clinical trial
- Challenges: Chemical validity, synthesizability, off-target effects — AI proposes but lab validates

More Key Applications of Generative AI

Audio & Music Generation

- Music Generation: MusicLM, Suno, Udio — text to full song with vocals
- Speech Synthesis: Text-to-Speech using neural vocoders (WaveNet, VITS, ElevenLabs)
- Voice Cloning: Clone voice from 5 seconds of audio — accessibility but also deepfake risk
- Audio Inpainting: Fill missing audio segments, de-noise recordings
- Model Architecture: AudioLDM uses LDM on mel-spectrograms (2D audio representations)
- MusicGen (Meta): Transformer-based music generation with conditioning on melody/genre
- Challenge: Rhythm, harmony, long-form coherence — music has complex hierarchical structure

Science & Engineering

- Weather Forecasting: GraphCast, Pangu-Weather — 10-day forecasts, outperform traditional models
- Climate Modeling: Generate ensemble simulations for climate scenarios
- Materials Science: GNoME (Google DeepMind) — discovered 2.2M new crystal structures
- Code Synthesis: Devin, SWE-bench — autonomous software engineering agents
- Medical Imaging: Synthetic MRI/CT for training, rare disease augmentation
- Robotics: RT-2, Genie — generative models for robot policy and environment simulation
- Finance: Synthetic financial data generation for backtesting without data leakage

Evaluating Generative Models — Key Metrics



- FID (Fréchet Inception Distance): Compares feature distribution of real vs generated images using InceptionNet — lower is better
- IS (Inception Score): Measures quality AND diversity — but biased toward ImageNet classes
- Precision & Recall (for GANs): Precision = sample quality, Recall = coverage of data distribution
- CLIP Score: Measures text-image alignment — how well image matches the text prompt
- Human Evaluation: MOS (Mean Opinion Score) — gold standard but expensive and slow
- BLEU, ROUGE, BERTScore: For text generation — measures n-gram overlap and semantic similarity
- Log-likelihood / ELBO: For VAEs and flow models — directly measures probability of test data
- Challenge: No single metric captures all aspects — quality, diversity, alignment, novelty

Ethics, Challenges & Future Directions

Responsible AI development and what comes next

Ethical Challenges in Generative AI



- Deepfakes: Synthetic media of real people — political manipulation, non-consensual content, fraud
- Copyright & IP: Training on copyrighted data without consent — ongoing lawsuits (NYT vs OpenAI)
- Bias Amplification: Models trained on biased internet data perpetuate and amplify societal biases
- Misinformation: AI-generated text/images can produce believable false news at massive scale
- Environmental Impact: Training GPT-4 estimated at ~500 tons CO₂eq — carbon footprint concerns
- Job Displacement: Creative workers (artists, writers, coders) face automation of significant tasks
- Surveillance: Generative models for voice/face synthesis enable impersonation and fraud
- Regulation: EU AI Act (2024), US Executive Order — first major regulatory frameworks for AI

AI Safety & Alignment in Generative AI



- Alignment Problem: Ensuring AI systems behave according to human values and intentions
- RLHF (Reinforcement Learning from Human Feedback): Train reward model on human preferences, fine-tune LLM — used in ChatGPT, Claude, Gemini
- Constitutional AI (Anthropic): AI critiques and revises its own outputs using a set of principles
- Red-teaming: Deliberately probing models for harmful outputs — mandatory before deployment
- Content Safety: NSFW classifiers, prompt filtering, output moderation — layered defense
- Watermarking: C2PA standard, StegaStamp — mark AI-generated content for traceability
- Hallucination Mitigation: RAG, chain-of-thought reasoning, uncertainty quantification
- Model Cards & Datasheets: Transparency about training data, intended use, limitations



Text (LLMs)

GPT-4o, Claude 3.5 Sonnet, Gemini Ultra,
LLaMA 3.1

Image

DALL-E 3, Midjourney v6, FLUX.1, Adobe
Firefly 3

Video

Sora (OpenAI), Lumiere (Google),
RunwayML Gen-3

Audio/Music

ElevenLabs, Suno v3, Meta MusicGen

3D / World

LucidDreamer, Genie (DeepMind),
DreamGaussian

Science

AlphaFold 3, GNoME, WeatherBench2
(GraphCast)

Future Directions in Generative AI



- Multimodal Foundation Models: Unified models handling text, image, audio, video, 3D simultaneously
- World Models: AI systems that model physical world dynamics — foundation for robotics and AGI
- Agentic AI: GenAI-powered agents that plan, use tools, browse web, write & run code autonomously
- Personalization: Models fine-tuned/adapted to individual users at inference time (DreamBooth, LoRA)
- Efficiency: Smaller but capable models (phi-3, Mistral, Gemma) — on-device generation on smartphones
- Scientific Discovery: AI-driven hypothesis generation, experiment design, literature synthesis
- 3D & 4D Generation: Consistent 3D world generation for gaming, AR/VR, simulation
- Regulation & Governance: International AI safety standards, audit frameworks, content provenance (C2PA)

Key Formulas — Exam Reference Sheet

Gaussian PDF

$$f(x) = (1/\sigma\sqrt{2\pi}) \exp(-(x-\mu)^2/2\sigma^2)$$

VAE ELBO Loss

$$L = E[\log p(x|z)] - KL(q(z|x) \parallel p(z))$$

GAN Objective

$$\min_G \max_D E[\log D(x)] + E[\log(1-D(G(z)))]$$

Diffusion Loss

$$L = E[\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]$$

KL Divergence

$$D_{KL}(P \parallel Q) = \sum P(x) \log[P(x)/Q(x)]$$

VAE KL Term

$$KL = -\frac{1}{2} \sum (1 + \log \sigma^2 - \mu^2 - \sigma^2)$$

Diffusion Forward

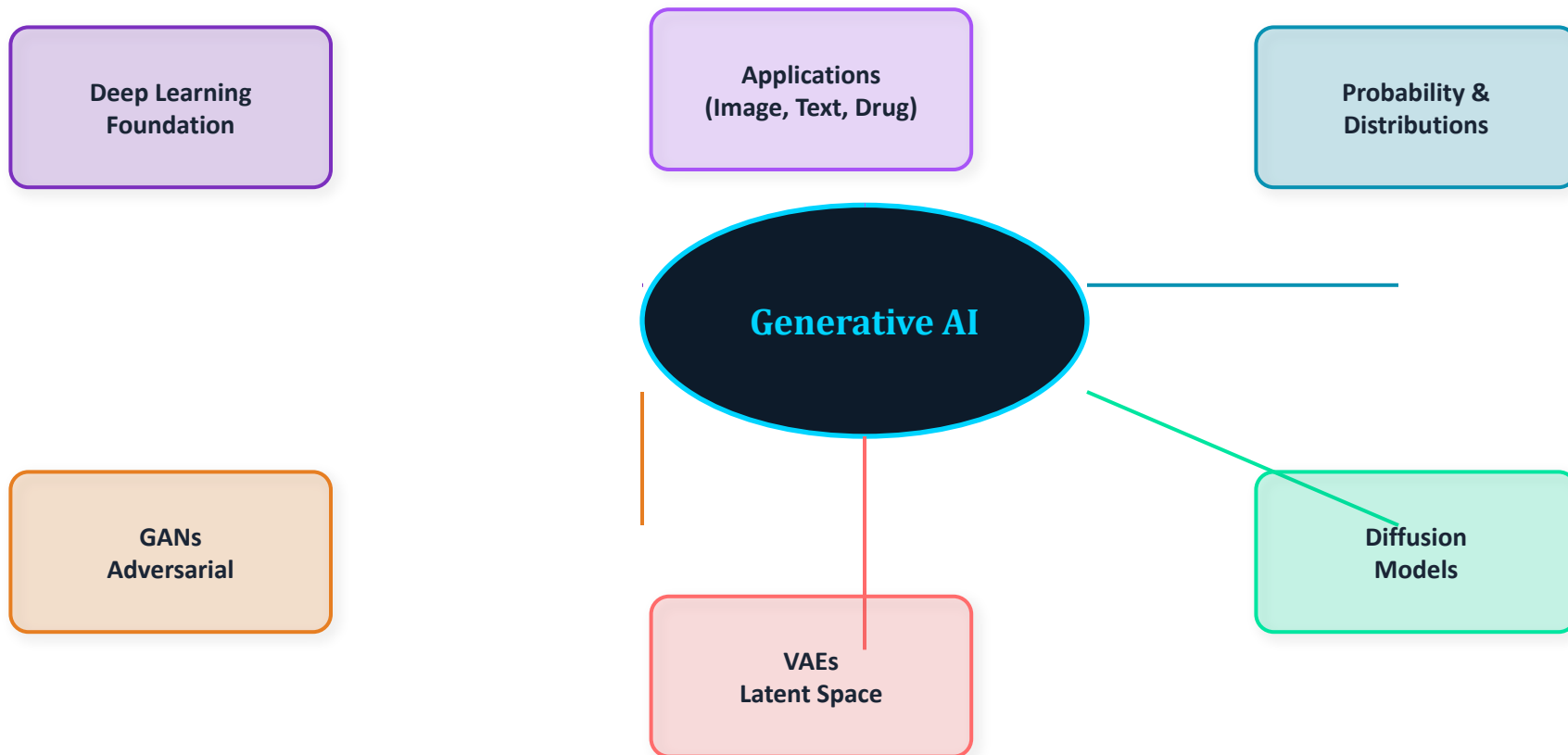
$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{(1-\alpha_t)} \epsilon, \quad \epsilon \sim N(0, I)$$

CFG Sampling

$$\hat{\epsilon} = \epsilon_{\text{uncond}} + w \cdot (\epsilon_{\text{cond}} - \epsilon_{\text{uncond}})$$



Unit 1 — Concept Map & Connections



Likely Exam Questions — Unit 1



- Q1: Explain the difference between discriminative and generative models with mathematical definitions
- Q2: Derive the VAE ELBO. Explain the role of each term and the reparameterization trick
- Q3: Explain the GAN minimax objective. What happens at Nash equilibrium?
- Q4: What is mode collapse in GANs? How does WGAN address this problem?
- Q5: Describe the forward and reverse process of DDPM. Write the training objective
- Q6: Compare GANs, VAEs, and Diffusion Models on: training stability, sample quality, latent space
- Q7: What is Classifier-Free Guidance? How does guidance scale affect text-to-image generation?
- Q8: Give 5 applications of generative AI in different domains with specific model examples

References & Further Reading



- Seminal Papers:
 - Goodfellow et al. (2014) — Generative Adversarial Networks
 - Kingma & Welling (2013) — Auto-Encoding Variational Bayes (VAE)
 - Ho et al. (2020) — Denoising Diffusion Probabilistic Models (DDPM)
- Books:
 - Deep Learning — Goodfellow, Bengio, Courville (free at deeplearningbook.org)
 - Pattern Recognition & Machine Learning — Bishop (probabilistic foundations)
- Online Courses:
 - Stanford CS236: Deep Generative Models (ermongroup.github.io/cs236)
 - fast.ai Practical Deep Learning for Coders (free)
 - Hugging Face Diffusion Models Course (free)

References & Further Reading



- Resources & Code:
 - Hugging Face Diffusers Library — production diffusion models in Python
 - PyTorch Tutorials — official deep learning examples
 - Papers With Code (paperswithcode.com) — SOTA results with implementations
- YouTube Channels:
 - Andrej Karpathy (Neural Networks: Zero to Hero), Yannic Kilcher, Two Minute Papers

In every lecture, Faculty may talk on any of the following:

- **DCS & GSFCU Portal:** Remind students to use DCS, talk about it's features and instruct all to visit GSFCU website
- **Discipline, Values and Ethics.**
- **GSFC University – My University:** Emphasize the importance of taking ownership of the University.
- **Respect & Courtesy:** Remind students to be respectful to peers, teachers, and staff. Encourage polite language and behavior.
- **Punctuality:** Emphasize the importance of arriving on time to class and completing assignments by deadlines.
- **Classroom Rules:** Reinforce basic classroom rules, such as keeping mobile phones on silent and maintaining a clean environment.
- **Safety:** Remind students about safety protocols, whether it's physical safety in the classroom or digital safety when using online resources.
- **Positive Attitude:** Encourage a positive mindset, cooperation and supporting one another.

GSFCU & Values

Respect & Courtesy

Font

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Dress
Appropriately



Respect
Personal Space



Be Mindful
of Your Words



Respect
Coworkers' Time



Keep Your
Workplace Clean



Socialise with
Your Peers



Use Your
Mobile Wisely



Take
Responsibility

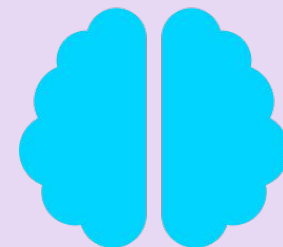


Thank You!

Questions & Discussion

Unit 1 Covered:

- AI, ML & Deep Learning foundations
- Generative Models: concepts & scope
- Probability Distributions & Sampling
- GANs: adversarial training framework
- VAEs: latent space & ELBO
- Diffusion Models: DDPM & guidance
- Applications across domains



*The
End*